

ign1t10n: macOS Installer Specification

One install: a multi-validator F1R3Node-Rust shard, its observer, the F1R3Gaze browser, and the F1R3Games portal, its playable games and F1R3Ink's relay, served to any web browser

Version 0.5 (draft for review)

Lucius Gregory Meredith

CEO, F1R3FLY.io, and Chief Science Officer, F1R3FLY Industries

9 October 2026

Abstract

This specification defines ign1t10n, an installer for macOS that leaves a person, a few minutes after double-clicking one file, with a working F1R3Node-Rust shard on their own machine (a bootstrap node, two to ten validators, and a read-only observer), the F1R3Gaze browser configured to deploy to that shard and read from it, and, new in this version, the F1R3Games portal and its games, installed on the shard and served on `http://localhost` so that an ordinary web browser (Safari, Chrome, Firefox) can sign on, launch new game instances, play them, and browse each game's gallery of plays. The portal is `f1r3games-service`, the Rust service of the F1R3Games repository, supervised by ign1t10n as one more process beside the nodes and Embers; each game client is served from an origin of its own, as the portal's frame model requires. Installation of the games on the chain (the portal's environment, each game's environment, and the registration of the games by a local stand-in for the F1R3FLY.io Cooperative) is automated, resumable and recorded in the manifest, replacing the eleven manual steps the first live installs of 5 and 6 October needed.

Version 0.5 brings the install up to the F1R3Games that now supports game play end to end: F1R3Pix, F1R3Beat and, new, F1R3Ink are each installed *playable*: client served, environment current, registered, and every administrative naming a game needs done on the chain. F1R3Ink's anonymous ink needs a *relay*, a server-side party other than the inker that deploys anonymous inks in shuffled batches; `f1r3games-service` now runs one, and ign1t10n supplies its keys, funds it, names it on the chain and keeps it named. From the menu bar the person can open the portal, start a new game of any registered game, or browse its gallery, in whatever web browser is the default.

The specification is grounded in ign1t10n branch `f1r3games` at 3456f69 (the implementation of version 0.4, packaged as `ign1t10n-0.2.0`), F1R3Games at d2381c6 (branch `f1r3ink-impl`: the portal with F1..F7 in tree, F1R3Pix, F1R3Beat, F1R3Ink and the relay), F1R3Node-Rust at fce422a, F1R3Gaze at f6ee26a, and the F1R3Games portal, F1R3Pix, F1R3Beat and F1R3Ink design documents.

Contents

1	What changed in version 0.5	4
2	What changed in version 0.4	4

3	Scope and conventions	5
3.1	What the installer delivers	5
3.2	Non-goals for this version	5
3.3	Pinned baselines	6
3.4	Conventions	6
4	Findings	6
4.1	The shard	6
4.2	F1R3Games	7
5	Architecture	10
5.1	Components	11
5.2	Principles	11
6	The distribution artifact	12
7	ign1t10n	12
7.1	Bundle layout	12
7.2	One binary, three roles	13
7.3	The launch agent	13
7.4	The state directory	13
7.5	Keys and Keychain items	13
8	Provisioning	14
8.1	Budgets	15
9	Supervision	15
9.1	Start	15
9.2	Health	16
9.3	Restarts	16
9.4	Stop	16
9.5	Port conflicts at start	16
9.6	Sleep and wake	16
10	Resizing the shard	16
11	F1R3Games	17
11.1	What runs	17
11.2	Origins and ports	17
11.3	Keys	18
11.4	Installation on the chain	18
11.5	Health	19
11.6	What the person does in the browser	19
11.7	F1R3Ink's relay	20
11.8	The F1R3Beat breeder (optional)	20
11.9	Upgrade	21
11.10	The F1R3Gaze rendition	21

12	User interface	21
12.1	Menu bar	21
12.2	The configuration panel	22
12.3	First-run window	22
13	Required changes	23
13.1	F1R3Gaze	23
13.2	F1R3Node-Rust	23
13.3	F1R3Games	23
14	Security model	24
15	Upgrade, reset and uninstall	25
15.1	Upgrade	25
15.2	Reset	25
15.3	Uninstall	25
16	Build and release pipeline	25
17	Testing and acceptance	26
18	Work packages and schedule	28
19	Obligations and decisions	28
19.1	Obligations	28
19.2	Resolved on 30 September 2026	29
19.3	Resolved on 7 October 2026	29
19.4	Open decisions	30
A	Configuration templates	31
B	The manifest	32

1 What changed in version 0.5

1. **F1R3Ink is installed and registered** with its web client (gallery renderers round and flag) at its own origin, :40703, reserved for it since version 0.4. Its environment rises to version 2: design v1 rewrote it, and installs of version 0.4 hold the old body at version 1 (Finding 24).
2. **F1R3Ink’s relay runs in the portal** (§11.7): a relay key and a handle secret join `games.secrets`; the portal’s configuration gains `[relay]`; jobs get a configuration of their own without it (Finding 22); stage G7 names the relay on the chain with F1R3Ink’s key, and `ign1t10n` keeps it named and funded. Opening the portal moves to G8.
3. **A game never registered here is upgraded without asking** (§11.9): no instance of it can exist, so replacing its environment discards nothing anyone made. Consent is still asked for every registered game and for the portal environment.
4. **Every game is reachable from the menu bar**: “New *game* game” and “*game* gallery” open `/games/<id>/launch` and `/games/<id>/gallery` in the default browser; `ctl games open [GAME [launch|gallery]]` does the same from a terminal.
5. **The decisions of version 0.4 are resolved as implemented** (§19.3), and version 0.4’s work packages F1–F7 have landed in the pinned F1R3Games revision. New: Findings 21–23, Principle 4, work packages F8–F10 and H10–H11, acceptance tests A24–A29, and decisions 18–23.
6. **Defects of the version 0.4 implementation corrected**: F1R3Beat’s key signs `setBreeder` but was never funded, so naming the breeder could not pay its phlo; it is now funded first, as F1R3Ink’s key is before `setRelay`.

2 What changed in version 0.4

1. **F1R3Games is a component of the install** (default on, like Embers): the portal service, the F1R3Pix and F1R3Beat web clients, the games’ environments on the shard, their registration, and a “F1R3Games” item in the menu bar that opens the portal in the person’s default browser. New: Findings 11–25, Principles 1–3, §11 (the component), stages G1–G7 (§11.4), work packages F1–F7 in the F1R3Games repository (§13.3) and H7–H9, R2, T2 here, acceptance tests A15–A23, and decisions 9–17.
2. **The decisions of 30 September are folded in** (§19.2): the separate bootstrap is kept; Apple silicon only; macOS 13.0 minimum; 10 000 000 F1R3 to the browser’s wallet; deploys go to a randomly selected active validator; timings tuned for a laptop; the node refuses HTTP requests carrying a foreign `Origin` (N5); Embers is an installation option, default on.
3. **What the first working build taught is adopted as specification** (from `docs/IMPLEMENTATION.md` at 46faa56): the supervisor is a classic per-user launch agent loaded with `launchctl bootstrap`, not an `SMAppService` agent; the genesis ceremony waits for $N_0 - 1$ approvals; all six ports and the network id go on each node’s command line; the bootstrap’s node id is read from its output; node logs go to standard output; templates are compiled into the binary; the archive is `macos-arm64`.

3 Scope and conventions

3.1 What the installer delivers

After a successful install on a supported Mac, with no further action beyond approving a background item in System Settings when macOS asks:

1. `/Applications/F1R3Gaze.app` and `/Applications/ign1t10n.app` are installed, signed and notarised.
2. A per-user launch agent, `io.f1r3fly.ign1t10n.supervisor`, is loaded and running. It supervises:
 - a *bootstrap* node, which coordinated the genesis ceremony and is every other node's rendezvous; it is not bonded and never proposes;
 - N *validators*, bonded at genesis or by a later resize, which propose and finalise blocks ($N = 2$ unless the person chose otherwise, $2 \leq N \leq 10$);
 - one *observer*, a read-only node that serves every read;
 - Embers, unless the person turned it off;
 - the *portal*, `f1r3games-service`, unless the person turned F1R3Games off.
3. Every key was generated on this machine at install time. The shard's genesis funds the person's active F1R3Gaze wallet and a local faucet; the faucet funds Embers, the portal and the local Cooperative.
4. F1R3Gaze deploys to a randomly selected active validator, reads from the observer, and is open in a window.
5. The F1R3Games portal environment and the five game environments are installed on the shard; F1R3Pix, F1R3Beat and F1R3Ink are registered and playable; F1R3Ink's relay runs in the portal and is named on the chain; the portal answers at `http://localhost:40700` and has been opened in the default browser, where the person can create a keystore, be funded by the faucet, launch a game, play it, and browse each game's gallery.
6. Every listening socket of every supervised process is bound to a loopback address.
7. The shard and the portal survive logout, sleep and reboot, and come back with the chain, the portal's origin and every browser keystore intact, at the size last chosen.

Throughout, “ N nodes” in the user interface means N validators; the bootstrap and the observer are always present and are shown separately. With the defaults, five `f1r3node` processes, Embers and the portal run.

3.2 Non-goals for this version

More than ten validators; joining a remote or public shard; exposing the local shard or the portal to the network (so invitations work for other browsers and other accounts on this Mac only); stake other than equal stakes; Intel Macs, Linux and Windows; Docker; running the F1R3Games portal *inside* F1R3Gaze (it executes no JavaScript; §11.10); web clients for F1R3SideChat and F1R3Skein, which do not exist yet; the F1R3Skein visionOS instrument; a relay for any game but F1R3Ink, and a relay run apart from the portal (Decision 19).

3.3 Pinned baselines

Source	Revision	Notes
ign1t10n	f1r3games 3456f69	@ the implementation of v0.4 (on main @ 46faa56, v0.3), packaged as ign1t10n-0.2.0; docs/IMPLEMENTATION.md records its decisions, deviations and known issues
F1R3Node-Rust	master @ fce422a	crate node v0.4.46; toolchain nightly-2026-02-09; topology reference docker/shard.yml
F1R3Gaze	main @ f6ee26a	release v0.1.0; crates gaze-shard, gaze-wallet reused by ign1t10n; renders HTML with Blitz and runs no JavaScript
Embers	28e50d4	bundled as Helpers/embers; its contracts do not parse on fce422a (known issue)
F1R3Games	f1r3ink-impl d2381c6	@ Portal/f1r3games-portal (Rust workspace: core, wallet, node, service, cli, games, rholint; React shell in web/); F1R3Pix/client, F1R3Beat/client, F1R3Ink/client; contains F1-F7 and the relay (F8); to be re-pinned to main once merged
Designs	5-8 October 2026	F1R3Games Portal design v0.1; F1R3Pix design v1; F1R3Beat design v2; F1R3Ink design v1

3.4 Conventions

MUST, MUST NOT, SHOULD, SHOULD NOT and MAY carry their RFC 2119 meanings. “The state directory” is ~/Library/Application Support/F1R3FLY/ign1t10n. “The profile” is F1R3Gaze’s profile directory. “The portal” is the running f1r3games-service process together with the web shell it serves; “the portal environment” is the games environment it installs on the shard; “a game environment” is one game’s contract. “The local Cooperative” is a key ign1t10n generates to stand in for the F1R3FLY.io Cooperative on this shard (Decision 10). “The relay” is F1R3Ink’s relay as f1r3games-service runs it; “naming the relay” is F1R3Ink’s setRelay, by which the holder of F1R3Ink’s key tells its environment which key may write anonymous stripes. A game is *playable* when Principle 4 holds for it. “An origin” is a scheme, host and port, as browsers define it. Ports are written for the default allocation.

4 Findings

These are the facts the design rests on. Findings 1–10 are carried from version 0.3, shortened where the implementation has since settled them; Findings 11–25 are new.

4.1 The shard

Finding 1 (The observer is not optional). Every endpoint that runs an exploratory deploy is served only by read-only nodes; a validator answers 400 readonly_node_required. That covers POST /api/explore-deploy, POST /api/estimate-cost, GET /api/registry/{uri} and GET /api/balance/{address} (docs/node/api-reference.md). F1R3Gaze, Embers and f1r3games-service all read there.

Finding 2 (The tested topology). docker/shard.yml runs a bootstrap started with --ceremony-master-mode, --required-signatures and --heartbeat-disabled, absent from bonds.txt; genesis validators started with --genesis-validator and --bootstrap pointing at it; and an ob-

server bootstrapping from it. `ign1t10n` runs that shape on loopback, generalised from three validators to N .

Finding 3 (Same host, distinct ports). On one Mac every node advertises `127.0.0.1` and differs by port. The node's port options have default values that override the configuration file, so all six ports go on the command line; the outgoing network id defaults to `testnet` unless `--network-id` is given (both found on the first real multi-node run).

Finding 4 (Approvals). The node requires strictly fewer ceremony approvals than genesis validators; `shard.yml` uses 2 of 3. `ign1t10n` passes $\max(N_0 - 1, 1)$.

Finding 5 (The shard can be resized while running). A node that was not at genesis joins by last-finalised-state fetch, is funded, submits a bond deploy to a *bonded* validator, and becomes active at the next epoch boundary (`docs/node/joining-a-network.md`); leaving is `PoS!("withdraw", ...)`, effective at the next boundary, with payout after the quarantine. `epoch-length` and `quarantine-length` are 10, so both take seconds to minutes locally. A bonded validator that does not propose is counted in consensus while contributing nothing.

Finding 6 (Deploys can be signed outside the node). `F1R3Gaze`'s `gaze-shard` produces deploys byte for byte as the node verifies them; `gaze-wallet` renders the vault transfer. `ign1t10n` uses both for faucet transfers, bond, withdraw and sweep deploys.

Finding 7 (The shipped development keys are public). `docker/.env.example` and the committed `docker/.env` carry the private keys of every node in the Docker topologies and control the genesis supply there. The `F1R3Games` repository likewise commits example keys under `Portal/f1r3games-portal/examples/local-shard/` (a service key, an environment key, a token secret and five game environment keys). No install may reuse any of them.

Finding 8 (How a key can reach a process). `--validator-private-key` exposes the key in the argument vector; `--validator-private-key-path` takes an encrypted PEM whose password comes from `F1R3NODE_VALIDATOR_PASSWORD`. `f1r3games-service` reads its keys only from plaintext key files named in its configuration (`service_key_file`, `env_key_file`, `token_secret_file`) and the game keys from a directory (`--keys`); it reads them once, at start.

Finding 9 (Bind addresses and origins). `api-server.host` is configurable; the protocol transport and discovery bind `0.0.0.0` until $N1$ lands. With $N5$, the node answers 403 to any HTTP request carrying an `Origin` header, so no page in a stock browser can use a local node directly; programs that send no `Origin` (`F1R3Gaze`, `Embers`, `f1r3games-service`) are unaffected.

Finding 10 (Lifecycle facts). The node handles `SIGTERM`. A joined node restarted on the same data directory reconnects in about a second; its identity is the TLS certificate in the data directory, so a data directory must never be deleted while peers remember it. During the ceremony the bootstrap cannot answer `/api/status`; its node id is read from what it prints at start.

4.2 F1R3Games

Finding 11 (The portal is one Rust service). `f1r3games-service` (`crate f1r3games-service, axum`) serves the API under `/api` and, as a fallback service, the built React shell from `static_dir`, on the single address in `listen` (default `127.0.0.1:8640`). It holds three secrets: the *service key*, which pays the environment deploys and the testnet faucet; the *environment key*, which signs

the portal environment's `insertSigned` registration and so fixes its registry URI; and a *token secret*. It deploys to one `validator_url` and reads at one `observer_url`; it never proposes (validators propose by heartbeat, which `ign1t10n` leaves on). Subcommands: `serve` (default; installs the environment first when `bootstrap_env` is true), `bootstrap`, `env-uri`, `keygen`, `games-keygen`, `games-install --keys DIR --version V, games-manifests --keys DIR --entry-base URL -out FILE`. The portal design placed this logic in an Embers domain; the implementation chose a dedicated service, and the implementation governs here: F1R3Games does not need Embers.

Finding 12 (The browser never reaches the node). The shell's wallet is Rust compiled to WebAssembly (`crates/wallet`); it signs in the page and hands signed deploys to `f1r3games-service (/api/prepare, /api/send)`; reads go through `/api/explore` and the GET routes. The page holds no node URL. N5 therefore stands without exception.

Finding 13 (Games run on origins of their own). The portal frames a game in a sandboxed `iframe` (`allow-scripts allow-same-origin allow-forms allow-pointer-lock allow-downloads`) and answers its host protocol by `postMessage`, accepting only messages whose `event.origin` is the origin of the game's registered entry and whose source is that frame (`web/src/core/host.ts`). `allow-same-origin` is safe only because the game is on another origin, and `GameFrame.tsx` refuses to frame a game served from the portal's own origin. A game client is a static Vite build (`base: "/"`) loaded as `<entry>?instance=<id>&portal=<origin>`; its gallery renderers are `<entry>preview/<kind>.html`. F1R3Pix (renderer `canvas`), F1R3Beat (pattern, `session`) and, at the pinned revision, F1R3Ink (round, `flag`) have clients; F1R3SideChat has only an environment; F1R3Skein's manifest lists `visionos` and `web` and has no web client.

Finding 14 (Browser state belongs to an origin). The keystore, contacts and the pinned environment URI live in IndexedDB under the portal's origin (`web/src/core/store.ts`). A passkey is created with no explicit relying-party id, so it is bound to the origin's host; WebAuthn refuses IP literals as relying-party ids, and `localhost` is a secure context in Safari, Chrome and Firefox. Hence the portal MUST be reached as `http://localhost:port`, and that origin MUST not change for the life of the install: at `127.0.0.1:port`, or at another port, the person meets an empty keystore and passkeys that do not work.

Finding 15 (Keys fix the environment URIs, and wallets pin them). The portal environment's URI is derived from the environment key; each game environment's URI from that game's key; every call template in a manifest embeds its game's URI, so the manifest's template hashes depend on the keys. At sign-on the shell pins the portal environment URI and afterwards refuses a service that serves another (`web/src/core/portal.ts`). Rotating the environment key therefore strands every keystore in every browser.

Finding 16 (Installation is an ordered sequence of deploys). On a new chain: fund the service key; install the portal environment (one deploy of up to `env_phlo_limit`, 50 M phlo by default) and wait until the `env.probe` template reports the version; install each game environment (`games-install`); write the manifests (`games-manifests`), which take a single `--entry-base` and produce `<base>/<id>`; register each with `games.register`, which the environment accepts only from the `coop_address` that was rendered into it at install. Registration is today done with the `f1r3games` CLI, which needs its own passphrase-protected keystore holding the Cooperative's key (`register-games FILE [--only ID]`).

Finding 17 (Raising an environment's version discards its state). Installing a higher `env_version` of the portal environment re-initialises its maps: profiles, instances, plays, invitations and sponsorships on that shard are not carried over (F1R3Games docs/IMPLEMENTATION.md). Replacing a

game environment's body likewise replaces its per-instance state. A change to a game's call templates changes the manifest hashes and requires re-registration; a change confined to a web client changes nothing on chain.

Finding 18 (The faucet pays from the service key). `POST /api/testnet/fund` transfers `faucet.amount` dust (default 10^8 , i.e. 1 F1R3) from the service key to an address the page names, when `faucet.enabled`; the shell calls it during sign-on so a new key can pay for its first deploy.

Finding 19 (What the first live installs needed by hand). Standing the portal up on an `ign1t10n` shard on 5 and 6 October took: building the service and shell (including `npm install --include=dev` and `npm's install-script` approval for `esbuild`); `keygen` and `games-keygen`; reading the shard's endpoints with `ctl env`; funding the service key and a stand-in Cooperative with `ctl fund`; choosing a stand-in Cooperative address; editing `f1r3games.toml`; `bootstrap` and waiting; `games-install`; `games-manifests`; importing the Cooperative key into the CLI and `register-games`; and serving the game clients from a second development server (`:5174`) so they had their own origin. Defects found only on the live shard (the `deploy-id` parse, multi-line templates, `validAfterBlockNumber` from the validator's sequence number, `Nil` dropped from tuples, escaped manifest sources, `allow-same-origin`, `.contains` on lists, a `/\ List` binder) are fixed at the pinned revision. Every manual step above becomes a stage of `ign1t10n` (§11.4); no developer tool is needed on the person's Mac.

Finding 20 (`localhost` has two addresses). `localhost` resolves to `::1` and `127.0.0.1`, and browsers try both. A service listening on one family leaves the other free for any program to bind, and that program then answers some of the browser's requests: the same hazard version 0.3's implementation met with Docker's `*:p` listeners (`ports::port_free`).

Finding 21 (F1R3Ink's anonymity needs a relay). On a public chain an ink is anonymous only if someone other than the inker deploys it (F1R3Ink design §8, D2, D10). `f1r3games-service` at `d2381c6` runs that relay when its configuration has `[relay] enabled = true`: `POST /api/relay/f1r3ink` accepts a request the shell signed with the player's key under the domain `f1r3games/relay/v1`, naming the game, the instance and the relay URL from the registered manifest; the relay checks the signature, that the URL is its own, that the signer plays the round, that the round allows anonymous ink and has at least its `anonMin` players (default five), and a per-player limit (30 requests an hour); it queues the ink under a *handle*, an HMAC of the relay's secret, and once every `window_blocks` blocks (three) deploys the window's inks as one shuffled `f1r3ink.relayInk`, signed by the relay key, which pays the `phlo` (at most 5 M `phlo` per batch of 64). `reveal` deploys `relayReveal` at once. The relay keeps nothing durable: a restart loses at most one window, which the inkers' clients report as never landed. The environment accepts `relayInk` and `relayReveal` only from the address F1R3Ink's own key named with `setRelay`; until then every relayed ink is refused. The relay learns who inks whom; it cannot read a sealed colour or forge an ink.

Finding 22 (The service loads the relay's keys for every subcommand). `f1r3games-service` reads `F1R3GAMES_RELAY_KEY` and `F1R3GAMES_RELAY_SECRET` (or the configured files) whenever the configuration it is given enables the relay, before it dispatches any subcommand, `bootstrap` and `games-install` included. A job run with the portal's configuration would therefore have to be handed the relay's keys, which no job needs.

Finding 23 (The manifest names the relay; registration does not compare it). `games-manifests --relay-base B` gives a game declaring the relay capability (F1R3Ink only) a manifest field `relay =`

B/<id>; the shell posts only there, and refuses when the manifest has none. `register-games` treats a registered game as current when its status, id, entry, environment URI, capabilities, renderers and template hashes match, but not `relay`: a manifest that differs only in its relay is not registered again.

Finding 24 (F1R3Ink’s environment was rewritten). F1R3Ink design v1 replaced the environment’s methods (`enter`, `tags`, `visibility`, `ink` with sealed inks, `veil`, `say`, `close`; the relay’s `setRelay`, `relayInk`, `relayReveal`; six reads). The version 0.4 bundle recorded F1R3Ink’s environment at version 1, and `ign1t10n` 0.2.0 installed the then-current body at version 1 on every chain, but registered nothing, F1R3Ink having no client: no instance of it exists on any `ign1t10n` shard. A game can be launched only once registered, so a game never registered has no state worth keeping.

Finding 25 (F1R3Gaze runs no JavaScript). F1R3Gaze renders HTML and CSS with Blitz and executes no JavaScript; behaviour is `f1r3lang`. The React shell therefore runs only in a stock browser. The shell is cut so that a later `f1r3lang` rendition can replace it (`docs/GAZE-MAPPING.md`); that needs capabilities F1R3Gaze lacks today (allowances, `invite`, `open`, `pay`, `frame`, per-site storage).

5 Architecture

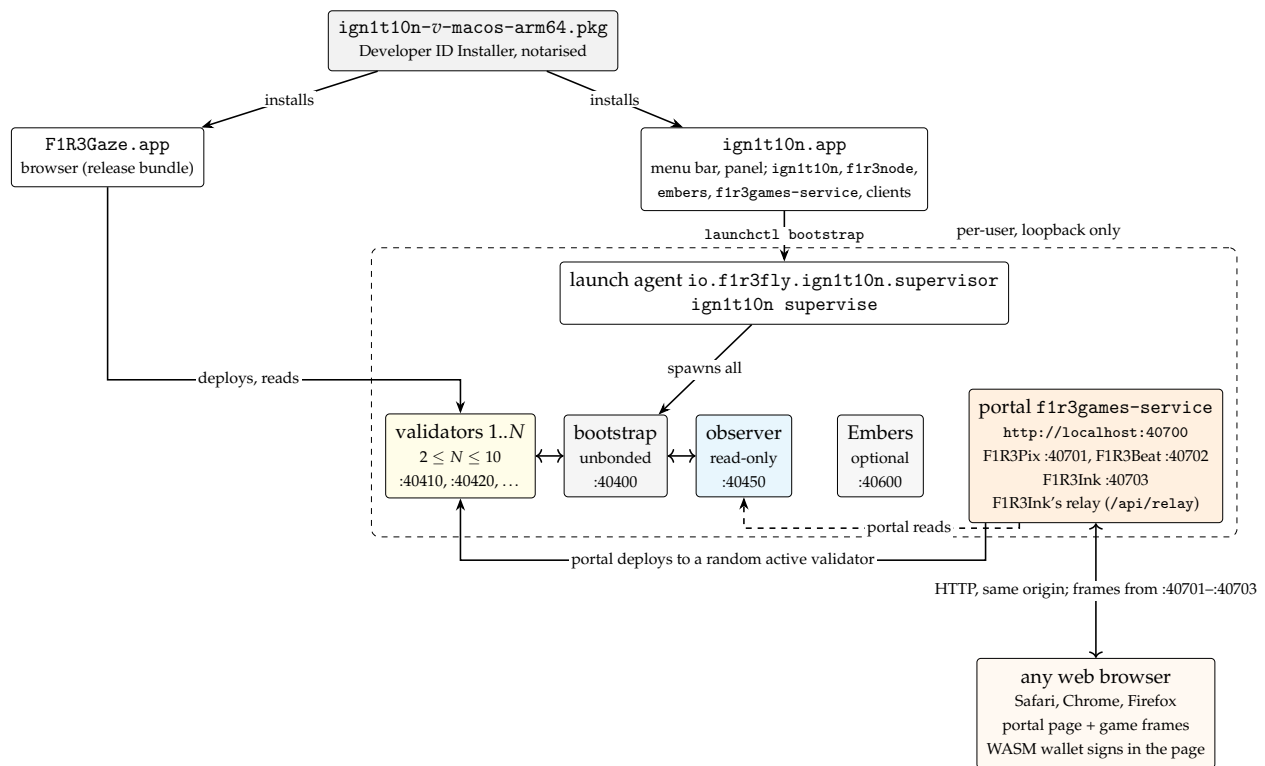


Figure 1: What the installer lays down and what runs afterwards, with the defaults. Nodes speak the peer protocol to one another through the bootstrap. The browser reaches only the portal; the portal alone reaches the nodes. The relay is part of the portal process and deploys, like the rest of it, to an active validator.

5.1 Components

Component	Role
F1R3Gaze.app	The F1R3Gaze release bundle, byte for byte.
ign1t10n.app	Menu-bar application and configuration panel; provisions, starts, stops, resizes, resets and reports on the shard and its services. Contains ign1t10n (one binary, three roles), f1r3node, embers, f1r3games-service, the f1r3games CLI, the portal shell and the game clients.
launch agent	io.f1r3fly.ign1t10n.supervisor; runs ign1t10n supervise, which owns every process below.
bootstrap, validators, observer	f1r3node run, as in version 0.3 (§9.1).
Embers	Optional; deploys to one active validator, reads at the observer.
portal	f1r3games-service serve: the F1R3Games API and shell at <code>http://localhost:40700</code> , each bundled game client at an origin of its own (F2), and F1R3Ink's relay at <code>/api/relay/f1r3ink</code> (F8); deploys to an active validator drawn per deploy (F7), reads at the observer.
F1R3Games jobs	One-shot runs of f1r3games-service for installation and registration (§11.4), of the f1r3games CLI to name F1R3Ink's relay (G7) and, if enabled, the F1R3Beat breeder (§11.8).
faucet	A key held by ign1t10n, never by a process, whose vault holds the undistributed genesis supply; funds joining validators, Embers, the portal, the local Cooperative and "Fund a wallet".
local Cooperative	A key held by ign1t10n; the address rendered into the portal environment as <code>coop_address</code> ; signs only <code>games.register</code> and <code>games.retire</code> . ign1t10n also runs, as the Cooperative's stand-in, the relay the F1R3Ink design assigns to the Cooperative (D10).

5.2 Principles

Version 0.3's principles stand: per-user and unprivileged at run time; loopback only; fresh secrets; one owner for the shard; a bonded validator runs; touch only what was created; idempotent, resumable operations. Version 0.4 adds three, and version 0.5 a fourth.

Principle 1 (Stable origins). The portal's origin, `http://localhost:port`, is fixed when it is first allocated and never changes automatically: not at start, not on a port conflict, not at reset, not at upgrade. Every browser keystore lives there (Finding 14). The same holds for each game's origin, since manifests on the chain name it.

Principle 2 (The browser reaches only the portal). No page served by ign1t10n is given a node URL, and no node accepts a browser's request (N5). Everything a page does on the chain goes through f1r3games-service.

Principle 3 (Keys outlive chains). The portal's environment key and the games' keys are kept across resets and upgrades, so the environment URIs, and with them every pinned keystore and every manifest hash, survive a new chain (Finding 15). They change only when the person explicitly asks to rotate them. The relay's key and handle secret are among them: handles are derived from the secret, so a kept secret keeps every inker's handle on a target stable across a restart of the portal.

Principle 4 (Listed means playable). A game is registered, and so listed in the portal, only when everything its play needs is in place: its client served at its origin; its environment at the bundled version; and every key-holder naming it requires done on the chain (F1R3Ink’s relay; F1R3Beat’s breeder when the breeder is on). What is not yet in place degrades F1R3Games with a reason and is retried; it never leaves a listed game that silently cannot do what its launch page offers.

6 The distribution artifact

Requirement 1 (Product archive). The release artifact **MUST** be a flat product archive, `ign1t10n-v-macos-arm64.pkg`, built with `productbuild` from two component packages (`io.f1r3fly.f1r3gaze.pkg`, `io.f1r3fly.ign1t10n.pkg`), signed with a *Developer ID Installer* certificate, notarised and stapled. Each component **MUST** install its bundle to `/Applications` with bundle-version checking on, so an installed newer F1R3Gaze is not downgraded.

Requirement 2 (The post-install hook does one thing). The `postinstall` script of the `ign1t10n` component package (`io.f1r3fly.ign1t10n.pkg`) **MUST** do nothing but open `ign1t10n` in the console user’s session with `--first-run`, and **MUST** exit 0 even if that fails. It **MUST NOT** create files in the user’s home, load agents or generate keys. (Known issue at 46faa56: nothing is visible between the installer’s Close and `ign1t10n`’s window; `ign1t10n` **SHOULD** show its window before it restarts an existing supervisor.)

Requirement 3 (Platforms). Apple silicon only; every Mach-O is `arm64`; minimum macOS 13.0 (resolved decisions 2 and 3).

Requirement 4 (No dynamic dependencies outside the system). For every Mach-O in the archive, including `f1r3games-service` and `f1r3games`, `otool -L` **MUST** list only paths under `/usr/lib` and `/System/Library`.

Requirement 5 (No developer tools at install time). The portal shell, its WebAssembly wallet and the game clients **MUST** be shipped built. Nothing on the person’s Mac runs `npm`, `cargo`, `wasm-bindgen` or `Vite`.

7 ign1t10n

7.1 Bundle layout

```
ign1t10n.app/Contents/
  Info.plist                io.f1r3fly.ign1t10n, LSUIElement, arm64, min 13.0
  MacOS/ign1t10n            menu bar (default), `supervise`, `ctl`
  Helpers/f1r3node          f1r3node-rust `node` v0.4.46
  Helpers/embers            Embers 28e50d4
  Helpers/f1r3games-service the F1R3Games portal service          (new)
  Helpers/f1r3games         the F1R3Games CLI (Decision 16)      (new)
  Resources/
    ign1t10n.icns, StatusTemplate.pdf  F1R3FLY.io brand kit (firefly bug)
    compat.toml                        storage formats this node can open
    versions.toml                      what this release bundles (read at start)
    F1R3Gaze.app                      installed by S0 if F1R3Gaze is missing
    f1r3games/                        (new)
```

games.toml	per game: id, env version, template hashes, client
present	
portal/	web/dist: the shell, fonts, the wallet .wasm
games/f1r3pix/	F1R3Pix/client/dist (index.html, preview/canvas.html)
games/f1r3beat/	F1R3Beat/client/dist (index.html, preview/pattern.html,
session.html)	
games/f1r3ink/	F1R3Ink/client/dist (index.html, preview/round.html,
flag.html) (v0.5)	
uninstall.sh	

The clients are served read-only from inside the signed bundle, so an upgrade replaces them atomically and nothing in the state directory can alter them.

7.2 One binary, three roles

ign1t10n (Rust 1.91) is the menu bar and panel when run with no arguments, the supervisor as ign1t10n supervise, and a client of the control socket as ign1t10n ctl *verb*. Version 0.4 adds the verbs games on|off, games status [--json], games reinstall, games open, games env (prints the portal URL and the variables that point the f1r3games CLI at it) and games breeder on|off; version 0.5 adds games relay on|off and games open [GAME [launch|gallery]]. The control socket is run/control.sock, one JSON object per line, peer uid checked.

7.3 The launch agent

A classic per-user agent, ~/Library/LaunchAgents/io.f1r3fly.ign1t10n.supervisor.plist, naming the supervisor by absolute path and loaded with launchctl bootstrap gui/<uid>. ign1t10n rewrites and reloads it whenever the path or contents differ, and removes any SMAppService registration left by earlier builds. (Adopted from 46faa56: after a reinstall, launchd kept a stale background-task entry for the SMAppService agent and never started it again.) The agent still appears under Login Items. KeepAlive restarts only on an unsuccessful exit; ExitTimeOut covers the ordered shutdown.

7.4 The state directory

```
~/Library/Application Support/F1R3FLY/ign1t10n/    0700
shard.toml          the manifest (Appendix B), schema 4
keys/ genesis/ conf/ nodes/ run/ archive/      as in version 0.3
games/                                                     (new)
  f1r3games.toml      the portal's configuration (Appendix A.2), with [relay]
  f1r3games-jobs.toml the same without [relay], for jobs (v0.5)
  manifests/<id>.json  each game's manifest as registered
  cli/                F1R3GAMES_HOME for jobs that need the CLI
~/Library/Logs/F1R3FLY/ign1t10n/
ign1t10n.log, <node>.stdout.log, embers.stdout.log
portal.stdout.log, games-job.log                    (new)
```

7.5 Keys and Keychain items

Node and faucet keys are as in version 0.3: encrypted PKCS#8 PEMs, each with its own random password in a Keychain item of service io.f1r3fly.ign1t10n, account pem.name; Embers's se-

crets in `embers.secrets`. F1R3Games adds one item, `games.secrets`, a JSON object holding the service key, the environment key, the token secret, one environment key per game, the local Cooperative’s key, the breeder’s key and, since version 0.5, the relay’s key and its 32-byte handle secret (created on first use when an older install’s item lacks them). Its access control list is `ign1t10n`’s designated requirement. A key reaches a F1R3Games process only in that process’s own environment, and only the keys the process needs (§11.3).

8 Provisioning

Provisioning runs in the menu-bar process, as the user, when `shard.toml` is absent or records an incomplete stage. Stages S0–S9 are version 0.3’s, as implemented; the first-run window’s “Advanced” disclosure now offers the validator count (default 2), “Bundle Embers” (default on), “Install F1R3Games” (default on) and “Open F1R3Games when ready” (default on).

S0 preflight

macOS ≥ 13 ; Apple silicon; F1R3Gaze present (else install the bundled copy); memory and disk for $N_0 + 2$ nodes plus Embers and the portal (§8.1); the state directory creatable.

S1 ports

node blocks of six as in version 0.3 (bootstrap 40400, validators 40410–40440 then 40460–40510, observer 40450), Embers 40600, and, when F1R3Games is chosen, the portal and game ports of §11.2. A port is free only if it binds *and* nothing answers a connection to it, on `127.0.0.1` and, for the portal and games, on `::1`.

S2 the paying wallet

F1R3Gaze’s active wallet, or a new one labelled “Local shard”.

S3 keys

bootstrap, validators $1..N_0$, faucet; Embers secrets if chosen; `games.secrets` if chosen and absent (stage G1).

S4 genesis files

`bonds.txt` with equal stakes of 1 000; `wallets.txt`: 10 000 000 F1R3 to the browser’s wallet, 1 000 F1R3 to each validator, the rest of a 500 000 000 F1R3 supply to the faucet.

S5 node configurations

`common.conf` and one file per node (Appendix A.1).

S6 load the agent

§7.

S7 genesis

bootstrap, then validators with `--genesis-validator`, then the observer; record the genesis hash, node version and storage format.

S8 configure F1R3Gaze

the managed block or `settings.d` drop-in, with a randomly selected active validator.

S9 open F1R3Gaze

after the observer is ready.

When F1R3Games is chosen, the supervisor continues with stages G1–G8 (§11.4) once the shard is Running, and the first-run window lists them below S9. A failure in G1–G6 leaves the shard

Running and F1R3Games Failed; a failure in G7 leaves F1R3Games Degraded (the games play; only anonymous ink waits); neither undoes S0–S9.

8.1 Budgets

Targets on an M-series Mac with $N_0 = 2$: S0–S6 under 10 s of machine time; genesis under 120 s; observer join under 180 s; G1–G8 under 200 s (six environment deploys, three registrations and setRelay, each waited to finalisation, about one block each); whole first run under seven minutes. $M_{\min}$ and $D_{\min}$ add the measured footprint of Embers and of the portal (a placeholder of 150 MB each until T1 measures it). Plays are stored on chain, so disk grows with play: T2 measures the growth per published F1R3Pix canvas, F1R3Beat session and F1R3Ink round across the $N + 2$ data directories, and the panel shows it.

9 Supervision

9.1 Start

1. Read shard.toml; refuse to start if the node's storage format is not in compat.toml; probe every recorded port; on conflict go to §9.5.
2. Start the bootstrap; read its node id from its output.
3. Start every validator whose recorded state runs, in parallel; wait for each /api/ready.
4. Start the observer; wait for /api/ready.
5. If Embers is on: fund it once from the faucet; draw a random active validator; start it; wait for /api/service/ready.
6. Redraw F1R3Gaze's validator. Report Running.
7. If F1R3Games is on: run any incomplete G-stage or upgrade step (§11.9); draw a random active validator for the portal; start the portal; wait for readiness (§11.5). Report F1R3Games Running.
8. Resume an interrupted resize.

Node argument vectors (Appendix A.1 has the files they name):

```
# every node
f1r3node run --config-file "$STATE/conf/<node>.conf" \
--protocol-port B --discovery-port B+4 --api-port-grpc-external B+1 \
--api-port-grpc-internal B+2 --api-port-http B+3 --api-port-admin-http B+5 \
--network-id "$NETWORK_ID" --host 127.0.0.1 --no-upnp --allow-private-addresses \
--native-token-name F1R3CAP --native-token-symbol F1R3 --native-token-decimals 8
# bootstrap adds
--validator-private-key-path "$STATE/keys/bootstrap.pem" \
--required-signatures $(max(N0-1,1)) --ceremony-master-mode --heartbeat-disabled
# validator k adds
--bootstrap "rnode://$BOOT_ID@127.0.0.1?protocol=40400&discovery=40404" \
--validator-private-key-path "$STATE/keys/validator-k.pem" [--genesis-validator]
# observer adds
--bootstrap "rnode://$BOOT_ID@127.0.0.1?protocol=40400&discovery=40404" --heartbeat-disabled
```

The portal runs as `f1r3games-service -c "$STATE/games/f1r3games.toml" serve` with the reduced environment (HOME, PATH, TMPDIR, LANG) plus its keys (§11.3) and `RUST_LOG=info`; output goes to `portal.stdout.log`.

9.2 Health

Every 10 s the supervisor reads `/api/status` from every node, Embers's readiness and the portal's (§11.5). The shard's status machine is unchanged: `Stopped` $\rightarrow$ `Starting(stage)` $\rightarrow$ `Running` $\rightleftharpoons$ `Degraded(reason)`, plus `Resizing(step)` and `Failed(reason)`. F1R3Games has its own: `Off`, `Installing(stage)`, `Running`, `Degraded(reason)`, `Failed(reason)`, `Waiting` (the shard is not Running). The menu shows both.

9.3 Restarts

A node that exits unexpectedly is restarted with exponential back-off (1, 2, 4, ..., 60 s); five unexpected exits of one node in five minutes stop the whole shard in `Failed` rather than leave a bonded validator down. Embers and the portal are restarted the same way, but five exits in five minutes put only that service in `Failed`; the shard keeps running.

9.4 Stop

On `SIGTERM`: the portal (and any F1R3Games job) and Embers, waiting up to 30 s; the observer, 30 s; every validator at once, 60 s; the bootstrap, 30 s; `SIGKILL` whatever remains. "Stop shard" stops everything but the supervisor; `uninstall` and `Shutdown` also make it exit 0.

9.5 Port conflicts at start

If a node's or Embers's recorded port is taken, that component is `Failed`, naming the port and its holder (lsyf), and the menu offers "Choose new ports" for every clashing node at once. The portal and game ports are *not* offered a silent move (Principle 1): F1R3Games goes `Failed` naming the holder and asks the person to stop it. "Move F1R3Games to a new address..." exists in the panel behind a dialog that says browser keystores at the old address will not follow, offers to open the portal's Wallet page so each key can be exported first, and then re-registers the games under their new origins (G6).

9.6 Sleep and wake

The processes run under a launch agent, so App Nap does not apply. The supervisor holds a `PreventUserIdleSystemSleep` assertion only during provisioning, resizes and G-stages.

10 Resizing the shard

Version 0.3's procedure stands: slots grow (keyed, funded, joining, bonding, active) one after another and shrink (retiring, left, paid, absent) from the highest; the node is stopped only after it has left the active set; a failed step stops at that slot and offers `Retry` or `Undo`; "Start a new shard with *M* validators" is a reset. One addition: before a validator withdraws, *every* client pointed at it is moved to another active validator: F1R3Gaze's setting, Embers, and now the portal, which is restarted with the new `validator_url` (a restart costs about a second; deploys a page sent in that second fail and are retried by the page). With F7 the portal takes the whole list and needs no restart.

11 F1R3Games

11.1 What runs

Process	Command	Purpose
portal	<code>f1r3games-service serve</code>	API and shell at <code>http://localhost:40700</code> ; F1R3Pix at <code>:40701</code> , F1R3Beat at <code>:40702</code> , F1R3Ink at <code>:40703</code> (F2); faucet on; F1R3Ink's relay at <code>/api/relay/f1r3ink</code> unless switched off (F8)
install job	<code>bootstrap; games-install</code>	G4, G5; upgrades
register job	<code>games-manifests --relay-base; register-games</code> (F3)	G6; re-registration
relay job	<code>f1r3games ink set-relay</code> (F8)	G7; again after a reset or a new F1R3Ink environment
breeder job	<code>f1r3games beat ...</code> (F6)	optional, daily (§11.8)

Only the portal is long-lived; the relay lives inside it (Decision 19). Jobs are run by the supervisor one at a time, with the jobs' configuration `games/f1r3games-jobs.toml`, which is the portal's without `[relay]` (Finding 22), logged to `games-job.log`, and recorded in `shard.toml` before and after, so an interrupted job is re-run (every job is idempotent: each probes the chain before it deploys, Finding 16; naming the relay twice names the same key).

11.2 Origins and ports

Origin	Port	Serves	Manifest entry
portal	40700	<code>/api</code> , the shell	—
F1R3Pix	40701	<code>/f1r3pix/</code>	<code>http://localhost:40701/f1r3pix/</code>
F1R3Beat	40702	<code>/f1r3beat/</code>	<code>http://localhost:40702/f1r3beat/</code>
F1R3Ink	40703	<code>/f1r3ink/</code>	<code>http://localhost:40703/f1r3ink/</code>
SideChat, Skein	40704–40705	reserved	not yet registered

Requirement 6 (The host is `localhost`). Every URL `ign1t10n` writes for the portal or a game (manifest entries, the relay's `base_url` and the manifests' `relay`, `portal_base_url`, the menu's "Open F1R3Games", "New ... game" and "... gallery", "Copy F1R3Games address") MUST use the host `localhost`. The portal MUST redirect a request whose `Host` is `127.0.0.1:p` or `:::1:p` to `http://localhost:p` (308) and MUST refuse any other `Host` (421), which also defeats DNS rebinding (F4).

Requirement 7 (Both loopback families). The portal and every game origin MUST listen on `127.0.0.1` and on `:::1` at the same port (F4); a port is allocated only if it is free on both (Finding 20). Nothing listens on any other address.

Requirement 8 (The relay is same-origin with the shell). The relay's URL MUST be the portal's origin followed by `/api/relay/f1r3ink`: the shell posts to it from that origin, so no CORS origin is needed and none is configured (`cors_origins = []`). A game client never reaches the relay; it asks the shell through the host protocol's relay capability.

Requirement 9 (One origin per game). Each bundled game **MUST** be served from its own port, serving only its own directory, with `Content-Security-Policy: frame-ancestors http://localhost:40700` and `X-Content-Type-Options: nosniff` (Decision 12). Games then share neither storage nor a `postMessage` origin.

Ports are allocated once, at G2, from the plan or else the first free decade in 41700–49990, and recorded; they are never moved except by the person (§9.5). The block 40700–40709 is outside the ports a developer’s F1R3Games checkout uses (8640, 5173, 5174), so a development portal and the installed one run side by side against the same shard (Decision 9).

11.3 Keys

Key	Funded with	Given to
service key	1 000 000 F1R3	portal (F1R3GAMES_SERVICE_KEY); install job; register job (pays)
environment key	—	portal (F1R3GAMES_ENV_KEY); install job
token secret	—	portal (F1R3GAMES_TOKEN_SECRET)
game keys (5)	F1R3Ink’s and F1R3Beat’s 10 F1R3, before they sign	install job and register job (F1R3GAMES_GAME_KEY_<ID>); F1R3Ink’s to the relay job, for <code>setRelay</code> ; F1R3Beat’s to the breeder job once, for <code>setBreeder</code>
local Cooperative	1 000 F1R3	register job only (F1R3GAMES_COOP_KEY)
breeder	1 000 F1R3	breeder job only
relay key	1 000 F1R3, topped up below 100	portal only, when the relay runs (F1R3GAMES_RELAY_KEY)
relay handle secret	—	portal only, when the relay runs (F1R3GAMES_RELAY_SECRET)

All eight kinds live in `games.secrets` and reach a process only in its environment (F1, which has landed); the CLI receives the one key it signs with as `F1R3GAMES_KEY`. No key is written to a file, an argument vector or a configuration.

11.4 Installation on the chain

Each stage is recorded in `shard.toml` under `[games.stages]` and resumes at its first incomplete step.

G1 secrets

Generate `games.secrets` if it is absent; otherwise keep it (Principle 3). Record the service, Cooperative and breeder addresses and every environment URI (`f1r3games-service env-uri` and the game-key derivation).

G2 ports

Allocate the portal’s and the bundled games’ ports (§11.2) unless already recorded.

G3 fund

Faucet transfers to the service key, the local Cooperative, (if the breeder is on) the breeder and (if the relay runs) the relay key and F1R3Ink’s key; wait until each is finalised. Skipped for an

address whose balance already covers its amount; re-checked for the relay whenever it comes on or F1R3Ink arrives after G3.

G4 configure and install the portal environment

Render `games/f1r3games.toml` (Appendix A.2) with a randomly selected active validator; run the install job's bootstrap; wait until the `env.probe` template at the observer reports `env-version` (F5's status `--json` reports it directly).

G5 install the game environments

`games-install --version V` with the versions from `Resources/f1r3games/games.toml`; wait until each game's probe reports its version. All five environments are installed, so registering a game whose client arrives later needs no deploy but the registration, unless its environment version rose meanwhile, which G5 then applies without asking (§11.9).

G6 register

For each game whose client is bundled: render its manifest with its own origin as entry (F3) and, for F1R3Ink, `--relay-base http://localhost:40700/api/relay` (F8); compare its SHA-256 with the one recorded in `shard.toml`; if the game is not registered or the hash differs, sign `games.register` with the local Cooperative's key and wait until `/api/games` lists it as given. Save the manifests. Then start the portal (§9.1) and wait for `/api/ready?games=...`

G7 name the relay

When the relay runs (§11.7) and `shard.toml` does not record it as named for F1R3Ink's current environment version: fund F1R3Ink's key if short; run `f1r3games -y ink set-relay <relay address>` with F1R3Ink's key as `F1R3GAMES_KEY` against the portal; wait until the deploy is finalised; record `relay_named = <F1R3Ink environment version>`. On failure, F1R3Games is Degraded ("F1R3Ink's relay is not named on the chain yet") and G7 is retried every 15 minutes.

G8 open

When the portal is ready and the person chose "Open F1R3Games when ready" at first run, open `http://localhost:40700/` in the default browser. Only at first run; later starts open nothing.

11.5 Health

The portal is *ready* when `GET /api/ready` (F5) answers 200, meaning: the node is reachable, the registered portal environment version equals the configured one, and every game recorded as registered is listed with the recorded hashes. Until F5, the supervisor checks `GET /api/env` (the URI equals the recorded one) and `GET /api/games` (each recorded game present). F1R3Games is Degraded when the portal is ready but the observer trails or a game's hashes differ from the bundle's (an upgrade is pending), or when the relay runs but is not named for F1R3Ink's current environment (G7 failed; retried every 15 minutes), and *Waiting* whenever the shard is not *Running*. Every 15 minutes the supervisor also reads the relay key's balance at the observer and, below 100 F1R3, tops it up to 1 000 F1R3 from the faucet.

11.6 What the person does in the browser

1. `http://localhost:40700/`: the feed. "Sign on" creates a keystore protected by a passphrase, and optionally a passkey, or imports a key file (the portable format F1R3Gaze and F1R3Sky use, so the person's F1R3Gaze wallet key can be brought in). The page pins the portal environment URI and asks the faucet to fund the new address with 100 F1R3 (Decision 14).

2. `/games`: F1R3Pix, F1R3Beat and F1R3Ink, from `games.register`. The menu bar’s “New game” and “game gallery” open the next two pages directly.
3. `/games/:id/launch`: a new instance, unlisted by default, with the game’s configuration (F1R3Pix: capacity 61, random seating; F1R3Beat: 4/4, two bars, sixteenths, capacity 32, 100bpm; F1R3Ink: a palette, decay, attribution and “Allow anonymous ink”, which needs the relay and at least anonMin players in the round). The launch sets an allowance so moves, and requests to the relay, are signed without a prompt; payments are always prompted.
4. `/instances/:id`: the lobby; invite (a link at localhost, useful to other browsers and other accounts on this Mac), start, play in the framed client.
5. `/games/:id/gallery`: plays by kind and day, each previewed by the game’s renderer from the game’s own origin (F1R3Ink: rounds, and portraits of single flags); `/plays/:id` opens one.

ign1t10n touches none of this; it guarantees only that the portal, its origins, the environments and the registrations are there.

11.7 F1R3Ink’s relay

The relay is on by default (Decision 18) and runs when it is on and F1R3Ink’s client is bundled. It is part of the portal: ign1t10n renders `[relay]` into `games/f1r3games.toml`,

```
[relay]
enabled      = true
base_url     = "http://localhost:40700/api/relay"  # what F1R3Ink's manifest names
window_blocks = 3                               # one batch per three blocks
per_hour     = 30                               # relayed requests per player
# key and handle secret: F1R3GAMES_RELAY_KEY, F1R3GAMES_RELAY_SECRET (portal only)
```

and starts the portal with the relay’s two secrets in its environment. The jobs’ configuration never has the table. Life cycle:

- *Naming*. G7 names the relay for F1R3Ink’s environment version and records the version. A reset (a new chain) or a new F1R3Ink environment (an upgrade, which re-initialises its state) makes the record stale, and G7 names it again; “Reinstall” names it again regardless, as a rotation of the F1R3Games keys (not yet implemented) must. Nothing else does: switching the relay off and on, a portal restart and a move to new ports keep it named.
- *Funding*. The relay pays for every batch and every reveal. It is funded with 1 000 F1R3 at G3 and topped up below 100 F1R3; at the limit of 30 requests a player an hour and at most 5×10^6 phlo (0.05 F1R3) per batch, that is thousands of batches before a top-up.
- *Off*. `ctl games relay off` (or the panel) restarts the portal without `[relay]` and without the relay’s keys. F1R3Ink stays registered and its manifest still names the relay’s URL (Decision 20); the portal answers that it runs no relay (404), and the shell reports anonymous ink as unavailable. Inks in the clear, sealed inks and everything else in F1R3Ink are unaffected. on restarts it with the relay, funding and naming it if needed.
- *What it knows*. The relay learns who inks whom in anonymous rounds; that is the trust the F1R3Ink design places in the Cooperative (D10). On the local shard the stand-in is ign1t10n, on the person’s own Mac, and the queue is in memory only.

11.8 The F1R3Beat breeder (optional)

F1R3Beat’s patterns breed in epochs run by a breeder key that F1R3Beat’s environment key has named (F1R3Beat design v2 §10). When “Run the F1R3Beat breeder daily” is on (Decision 15),

ign1t10n once funds F1R3Beat's game key, signs setBreeder with it (f1r3games beat set-breeder), launches a public nursery instance with the breeder's key, records its id, and then runs one epoch (f1r3games beat epoch, F6) once a day while the shard runs.

11.9 Upgrade

At start, the supervisor compares Resources/f1r3games/games.toml with what shard.toml records:

- a client-only change: nothing to do; the new bundle's files are served;
- a *registered* game's environment version rose: run G5 for that game at the bundled version and G6, *after consent*, since its live instances are replaced (Finding 17); until then the game stays registered at the old hashes and F1R3Games is Degraded ("F1R3Beat update waiting");
- a game *never registered on this chain* whose environment version rose: run G5 for it at the bundled version *without asking*, since no instance of it can exist (Finding 24), then G6 if its client is now bundled. This is how a version 0.4 install meets F1R3Ink: environment 1 → 2, then registration and G7, with no dialog;
- a game's template hashes changed with its environment version unchanged: G6 re-registers it (the manifest's hash differs), without consent: its environment and instances are untouched;
- the portal environment's version rose: the same, with a dialog saying that profiles, instances, plays, invitations and sponsorships on the local shard will be discarded; browser keystores are unaffected (the URI is unchanged).

11.10 The F1R3Gaze rendition

The shell's core/ layer is framework-free and talks to the world through eight capabilities, so that a f1r3lang page in F1R3Gaze can do the same (docs/GAZE-MAPPING.md). When F1R3Gaze gains those capabilities, ign1t10n will point F1R3Gaze at the portal's site (f1r3://f1r3fly/games) as well; nothing on the chain changes. Until then the menu opens the portal in the default browser and F1R3Gaze in its own window, and the two are independent.

12 User interface

12.1 Menu bar

A monochrome template image of the F1R3FLY.io firefly with a status dot: none for Running, amber for Starting, Resizing, Degraded and F1R3Games Installing, red for Failed, hollow for Stopped.

Item	Action
<i>Local shard: Running, 2 validators</i>	submenu: height, per-node readiness, observer lag, disk
<i>F1R3Games: Running, 3 games</i>	portal address, each game and its origin, the relay (running, named), last job, breeder
Open F1R3Games	open <code>http://localhost:40700/</code> in the default browser
New <i>game</i> game	for each registered game, its launch page <code>/games/<id>/launch</code> in the default browser
<i>game</i> gallery	for each registered game, its gallery <code>/games/<id>/gallery</code> in the default browser
Open F1R3Gaze	open <code>-b io.f1r3fly.f1r3gaze</code>
Start / Stop shard	through the control socket
Configure... (Command-comma)	the configuration panel
Fund a wallet...	faucet transfer to any address (a F1R3Gaze or portal key)
Copy endpoints	validators, observer, shard id, Embers, portal
Show logs	reveals <code>~/Library/Logs/F1R3FLY/ign1t10n</code>
Reset shard... / Uninstall...	\$15
Quit ign1t10n	quits the menu bar only

12.2 The configuration panel

The **Shard** and **General** panes are version 0.3's (validator stepper with footprint and measured tolerance, node table, Apply sheet; run at login, use the local shard in F1R3Gaze, ports, genesis amounts, Embers on/off). A third pane is new.

F1R3Games.

- **Install F1R3Games** on/off (`ctl games on|off`). Off stops the portal and frees its ports; it leaves the environments and registrations on the chain and the keys in the Keychain, so turning it on again needs no deploy.
- **Address:** `http://localhost:40700`, read-only, with "Open" and "Copy"; "Move to a new address..." behind the dialog of §9.5.
- **Games:** one row per game: origin, environment version, registered or not, client bundled or not, pending update with "Update..."
- **Faucet amount** for new portal keys (default 100 F1R3).
- **Run the F1R3Beat breeder daily** (default off) with the last epoch.
- **Run F1R3Ink's relay** (default on; `ctl games relay on|off`), with the relay's address, whether it is named on the chain, and its balance.
- **Reinstall** (re-runs G3–G6; nothing is deployed that is already there) and **Rotate F1R3Games keys...** (danger: new environment URIs; every browser keystore must be cleared, every game re-registered).

12.3 First-run window

F1R3FLY.io brand on a white surface (white-background logo; yellow and sky only as rules and progress), listing S0–S9 and then G1–G8, each with a spinner, check or error, and the "Advanced" disclosure of §8. It closes with a summary: shard id, validators, the funded F1R3Gaze wallet, the portal address with an "Open F1R3Games" button and one "Play" link per registered game, and where the logs are.

13 Required changes

13.1 F1R3Gaze

G1 (porcelain wallet commands), G2 (balances without Embers), G3 (`settings.d`), G4 (a validator list, one drawn per deploy) and G5 (a shard health page) are version 0.3's and `IMPLEMENTATION.md`'s, unchanged.

13.2 F1R3Node-Rust

N1 (bind addresses for transport and discovery), N2 (vendored OpenSSL and a macOS CI build), N3 (the loopback shard test with grow and shrink), N4 (storage-format number) and N5 (`api-server.reject-foreign-origin`) are unchanged (`docs/node-work-packages.md`). N5 does not conflict with F1R3Games (Finding 12).

13.3 F1R3Games

F1–F7 below are version 0.4's; all seven have landed and are in the pinned revision `d2381c6`. F8 landed with F1R3Ink's implementation; F9 and F10 are new requests.

Work package 1 (F1: keys from the environment). `f1r3games-service` reads `F1R3GAMES_SERVICE_KEY`, `F1R3GAMES_ENV_KEY`, `F1R3GAMES_TOKEN_SECRET`, `F1R3GAMES_GAME_KEY_<ID>`, `F1R3GAMES_COOP_KEY` and `F1R3GAMES_BREEDER_KEY` (hex) when set, before the configured files, and never logs them. Estimate: 0.25 engineer-week.

Work package 2 (F2: game origins served by the service). A repeated `[[origins]]` table: `id`, `listen` (a list), `dir`; each serves only `/<id>/` from `dir` (index fallback within it, no directory listing) with the headers of §11.2, on its own listeners in the same process. Estimate: 1 engineer-week.

Work package 3 (F3: headless registration and per-game entries). `games-manifests --entry ID=URL` (repeatable; `--entry-base` stays the default) and `f1r3games-service register-games FILE [--only ID]`, which signs `games.register` with `F1R3GAMES_COOP_KEY`, refuses when its address is not the environment's `coop_address`, skips a game already registered with the same manifest, and waits for finalisation. Estimate: 0.5 engineer-week.

Work package 4 (F4: dual-stack listeners and a Host allow-list). `listen` accepts a list; the default becomes `["127.0.0.1:P", "[::1]:P"]`. A configured `public_host` (`localhost:40700`) is the only Host served; loopback literals are redirected to it; anything else gets 421. The same for each `[[origins]]` entry. Estimate: 0.5 engineer-week.

Work package 5 (F5: readiness and status). `GET /api/ready` (§11.5) and `f1r3games-service status --json`: the portal environment's registered version, each game environment's, and each registered manifest's `id`, `entry` and template hashes. Estimate: 0.5 engineer-week.

Work package 6 (F6: a headless breeder (optional)). `f1r3games-service beat-set-breeder` and `beat-epoch --nursery ID`, signing with the keys of F1, computing the epoch as `f1r3games beat epoch` does. Estimate: 0.5 engineer-week.

Work package 7 (F7: several validators (optional)). `validator_urls`, one drawn per deploy, with a retry on another when one refuses, so the portal survives a validator's restart and needs no restart when one withdraws. Estimate: 0.25 engineer-week.

Work package 8 (F8: F1R3Ink’s relay (landed)). [relay] in the configuration; POST /api/relay/{game}; relay-keygen; games-manifests --relay-base; f1r3games ink set-relay; the shell’s relay capability (Finding 21). Landed at d2381c6.

Work package 9 (F9: compare the relay at registration). register::matches also compares relay, so a manifest that changes only its relay is registered again (Finding 23). With F9, ign1t10n could register F1R3Ink without relay while the relay is off, and the shell would say so before a round is launched, rather than at the first anonymous ink. Estimate: 0.1 engineer-week.

Work package 10 (F10: load the relay only to serve, and report it). Read the relay’s keys only for serve (Finding 22), so one configuration serves the portal and the jobs; and report the relay in GET /api/ready and status --json (its address, whether the chain names it, its queue and its last batch), so readiness can include it without ign1t10n recording the naming itself. Estimate: 0.25 engineer-week.

14 Security model

Threat	Protection
Reuse of public development keys	All keys generated on the Mac; the release pipeline fails if any public key from docker/.env.example or any address from F1R3Games’s examples/local-shard/ appears in an artifact or in shard.toml.
Another user on the Mac	State directory 0700; no key in any argument vector; passwords and F1R3Games keys only in the Keychain and in the owning process’s environment; control socket checks the peer uid. Loopback is shared between accounts: another user can open the portal and play, as any player can, and use its faucet; the faucet is the cost.
Hosts on the same network	Every listener on loopback (N1, api-server.host, F4).
Pages in other browsers or tabs	Cannot use a node (N5). Can reach the portal’s API only cross-origin: no CORS origin is configured, so responses are unreadable and JSON POSTs are refused at preflight. DNS rebinding is refused by the Host allow-list (F4).
A malicious or buggy game	Own origin and sandbox; the host answers only that frame and that origin; the wallet signs only the game’s registered play templates within the launch allowance, refuses fund-moving templates from a game, and prompts for payments (F1R3Games design).
Theft of a browser key-store	Encrypted under the passphrase (PBKDF2) or the passkey’s PRF output; it never leaves the browser profile; ign1t10n never reads it.
The local Cooperative key	Governs only the registry of this local shard; it is never given to the portal process.
F1R3Ink’s relay	Its key and handle secret are only in the portal’s environment, never in a job’s or a file; it can write only anonymous stripes in F1R3Ink rounds, for players whose signed requests it checked, at most 30 an hour each, and spends only its own balance (topped up to 1 000 F1R3). It learns who inks whom (D10); on this shard that is the person’s own Mac. Another account on the Mac can use it as any player can.
Tampered artifact	Developer ID signatures on every Mach-O and the archive; notarisation; hardened runtime with no entitlements; clients served from inside the signed bundle; GPG-signed SHA256SUMS.
Malware running as the user	Not defended.

15 Upgrade, reset and uninstall

15.1 Upgrade

A newer archive replaces both bundles. The supervisor compares the node's storage format (N4) with the recorded one and restarts every node on its data, or offers a new shard; then it runs §11.9 for F1R3Games.

15.2 Reset

"Reset shard..." archives the chain, node keys and configurations and re-provisions from S1 with the chosen N_0 . `games.secrets` is kept, so the environment URIs are the same; G3–G7 run again on the new chain (the relay is named again), and the portal's origin is the same. The dialog says that every deploy, registry entry, balance, profile, instance, play and gallery on the local shard will be gone; that F1R3Gaze wallets and browser keystores keep their keys and can sign on again at once; and that the faucet will fund them again. A checkbox "Also rotate the F1R3Games keys" (off) does what the panel's rotation does.

15.3 Uninstall

Stop everything; unload and remove the agent; after a second confirmation remove the state directory (optionally keeping archive/), the logs and every Keychain item of service `io.f1r3fly.ign1t10n` (including `games.secrets`); remove F1R3Gaze's managed block or drop-in; move `ign1t10n.app` to the Trash; `pkgutil --forget io.f1r3fly.ign1t10n.pkg`. F1R3Gaze and its wallets are left alone. `ign1t10n` cannot reach browser storage: the dialog says that keystores remain in each browser under `localhost:40700` until the person clears that site's data, and offers to open the portal's Wallet page first so keys can be exported.

16 Build and release pipeline

`versions.toml` gains a section, read by the release workflow and copied into the bundle:

```
[f1r3games]
repository = "https://github.com/F1R3FLY-io/F1R3Games"
revision   = "d2381c668e0ac4fddc1031c04b90c4f335c540c1" # branch f1r3ink-impl; main
              once merged
clients    = ["f1r3pix", "f1r3beat", "f1r3ink"]           # Decision 11
env_version = 1                                           # the portal environment
game_versions = { f1r3pix = 1, f1r3beat = 1, f1r3ink = 2, f1r3sidechat = 1, f1r3skein =
  1 }
has = ["F1", "F2", "F3", "F4", "F5", "F6", "F7", "F8"]
```

Job	Runner	Steps
node	macos-14	F1R3Node-Rust at the pinned revision, --features crypto/vendored-openssl, arm64; otool -L gate
embers	macos-14	Embers at the pinned revision, arm64
gaze	ubuntu	download and verify the pinned F1R3Gaze release
games (new)	macos-14	F1R3Games at the pinned revision: cargo test for the workspace and rholint; cargo build --release -p f1r3games-service -p f1r3games-cli, arm64; web/: npm ci, build the wallet to WebAssembly (wasm32-unknown-unknown, wasm-bindgen 0.2.129), npm test, npm run build; each listed client: npm ci, npm test, npm run build, and a renderer preview/<kind>.html for each gallery kind its game declares; write games.toml; otool -L gate
ign1t10n	macos-14	build, unit and end-to-end tests (the real f1r3games-service and f1r3games CLI against the stand-in node)
assemble	macos-14	build.sh with the F1R3Games artifacts; sign inside out (Helpers/*, MacOS/ign1t10n, the app); pkgbuild, productbuild, sign, notarise, staple
smoke	macos-14	headless provisioning with F1R3Games on; A2–A6, A11, A12, A15, A17, A23, A24, A25; A16 and A28 in headless Chromium and WebKit
publish	ubuntu	SHA256SUMS, GPG signature; draft release

17 Testing and acceptance

A1–A14 are version 0.3’s: clean install; round trip through F1R3Gaze; loopback only; no foreign libraries; no shipped secrets; persistence across reboot; port conflict with a Docker shard; crash recovery; upgrade; uninstall; grow in place; shrink in place; new shard at the maximum; bounds and interruption. A1 now also requires F1R3Games Running within seven minutes. New:

A15 F1R3Games installed.

After A1, `ctl games status --json` reports the portal ready, six environments at their bundled versions, and F1R3Pix, F1R3Beat and F1R3Ink registered with entries at `localhost:40701`, `:40702` and `:40703`.

A16 Playing in a stock browser.

Driven by Playwright in Chromium, WebKit and Firefox: sign on with a passphrase; the faucet funds the key; launch an F1R3Pix instance; seat, paint, publish a canvas; the gallery lists it and the renderer draws it; then the same for F1R3Beat with a pattern. Repeat sign-on with a passkey where the browser supports the PRF extension.

A17 Origins.

`http://127.0.0.1:40700/` redirects to `localhost`; a request with `Host: evil.example` gets 421; the portal refuses to frame a game whose entry is its own origin; game responses carry `frame-ancestors`; one game’s origin does not serve another’s files.

A18 Dual stack.

With a decoy listener on `[::1]:40700` before provisioning, G2 allocates another decade. With one started after provisioning, F1R3Games goes Failed naming the holder and does not move.

A19 Reset keeps identities.

A keystore created before “Reset shard” signs on afterwards without a pin error; the environment URIs are unchanged; G3–G6 ran.

A20 Upgrades.

A bundle whose only change is a client deploys nothing; one with a changed F1R3Beat template asks consent, installs version +1 and re-registers; F1R3Pix is untouched.

A21 Evacuation.

Shrinking away the portal's validator moves the portal first; a move sent during the shrink finalises.

A22 Off and on.

`ctl games off` frees ports 40700–40703; on restarts with no deploy and the same origin.

A23 Loopback.

`lsof` lists `f1r3games-service` only on `127.0.0.1` and `::1`.

A24 F1R3Ink registered with its relay.

`/api/games/f1r3ink` lists the entry at `:40703`, renderers round and flag, and `relay = http://localhost:40700/api/relay/f1r3ink`; its renderers are served with frame-ancestors.

A25 The relay runs and is named.

`ctl games status --json` reports the relay running and named; the node finalised exactly one `setRelay`; a malformed request to the relay is refused for its form or signature (not 404); the portal's configuration has `[relay]`, the jobs' has not; no relay key appears in any argument vector or file.

A26 Upgrade from version 0.4.

A shard provisioned by `ign1t10n 0.2.0` (F1R3Ink an unregistered environment at version 1), then started by version 0.5: F1R3Ink's environment is replaced at version 2 with no consent asked, registered, and the relay named; F1R3Pix and F1R3Beat are not registered again; nothing waits.

A27 Relay off and on.

`ctl games relay off` restarts the portal without the relay (its URL answers 404) and registers nothing; on restores it without naming it again. After a reset it is named again on the new chain.

A28 Anonymous ink in a stock browser.

Five keystores (five browser profiles) enter an F1R3Ink round launched with anonymous ink; each inks another anonymously; within one window plus finalisation every stripe shows its colour with no inker on chain, the relay's key paid the phlo, and a reveal attributes one. Then the same in a round with public flags, sealed inks and decay.

A29 The menu opens games.

"New F1R3Ink game" opens F1R3Ink's launch page and "F1R3Ink gallery" its gallery, in the default browser; neither item appears for a game that is not registered.

18 Work packages and schedule

WP	Deliverable	Est.	Status
N1–N5	node: bind addresses, macOS build, loopback shard test, storage format, origin guard	4.5	open
G1–G5	F1R3Gaze: porcelain, balances, <code>settings.d</code> , validator list, health page	2.5	open
H1–H6	ign1t10n: provisioning, supervision, UI, resize, panel, lifecycle	11	first working version
R1, T1	release workflow; acceptance and measurements	3.5	partly
F1–F5	F1R3Games: keys from environment, game origins, headless registration, dual stack and Host, readiness	2.75	new
F6, F7	F1R3Games: headless breeder; validator list (optional)	0.75	new
H7	ign1t10n: manifest schema 4, <code>games.secrets</code> , stages G1–G7, the portal role, jobs, health, evacuation, upgrade comparison	2.5	new
H8	ign1t10n: menu items, F1R3Games pane, first-run additions, opening the browser	1	new
H9	ign1t10n: reset, rotation, uninstall and upgrade dialogs for F1R3Games	1	new
R2	release: the games job, bundle layout, signing, smoke additions	1	new
T2	A15–A23; browser matrix; footprint and on-chain growth of plays	1.5	partly
F8	F1R3Games: F1R3Ink’s relay	—	landed (d2381c6)
F9, F10	F1R3Games: compare the relay at registration; relay only for <code>serve</code> , and reported	0.35	new
H10	ign1t10n: F1R3Ink’s client, the relay’s secrets, two configurations, G3 funding, G6 <code>--relay-base</code> , G7 naming, silent upgrades, <code>ctl games relay</code> , <code>ctl games open GAME</code> , top-ups, the breeder funding fix	1	implemented (spec-v0.5)
H11	ign1t10n: the menu’s launch and gallery items (implemented), the panel’s relay row and “Play” links in the first-run summary (open)	0.5	partly
T3	A24–A29; phlo per relayed batch on a real node	1	A24–A27, A29 against the stand-in node

The new work is about 10.5 engineer-weeks. F2, F3 and F5 are on its critical path: H7 needs game origins to register entries, headless registration to finish G6, and readiness to supervise. F1 and F4 can follow (H7 has an interim for F1). With one engineer on F1R3Games (F1–F7) and one on ign1t10n (H7–H9, R2), and T2 shared, the component fits in five to six calendar weeks, alongside the open version 0.3 work. Version 0.5 adds about two engineer-weeks, half of it already done; its critical path is T3’s A28 on a real node, which also discharges Obligation 1 for F1R3Ink.

19 Obligations and decisions

19.1 Obligations

Version 0.3’s obligations stand: the loopback shard works (N3); resizing works (N3); footprint (T1); measured fault tolerance (T1); PEM compatibility (discharged at 46f aa56). Added:

Obligation 1 (The environments behave on a node). Before H7 is merged, T2 must show, on an `ign1t10n` shard at the pinned node, the portal environment and the F1R3Pix, F1R3Beat and F1R3Ink environments doing what their designs say (for F1R3Ink, with the relay: `relayInk` batches, `relayReveal`, decay on block time), including the parts F1R3Games’s `IMPLEMENTATION.md` lists as unverified (the `TreeHashMap` handle, sponsorship escrow, `/\ Int` and `/\ String` patterns, `phlo` per move), and record `phlo` per move and per read so the allowance and capacity defaults rest on measurements.

Obligation 2 (Browsers). T2 must establish, for current Safari, Chrome and Firefox on macOS 13–26: that `http://localhost` is a secure context for WebAuthn, IndexedDB and WebAssembly; which support the PRF extension the passkey unlock needs (the passphrase works in all); and that the sandboxed game frames with `allow-same-origin` exchange messages with the portal. The first-run summary names any browser found wanting.

Obligation 3 (Origin stability). The end-to-end test must show that no path through start, port conflict, resize, reset, upgrade or F1R3Games off/on changes a recorded F1R3Games port.

Obligation 4 (Embers compatibility). Embers 28e50d4’s contracts do not parse at node `fce422a`. F1R3Games does not depend on Embers, but the release must not ship a pair that fails: pin a compatible Embers or default Embers off, and make the smoke job deploy its contracts.

19.2 Resolved on 30 September 2026

#	Decision	Resolution
1	Fold the bootstrap into validator 1	No: keep the separate bootstrap for version 1.
2	Intel	Apple silicon only.
3	Minimum macOS	13.0.
4	Genesis amounts	10 000 000 F1R3 to the browser’s wallet; validators 1 000 F1R3 fees and 1 000 stake; the rest of 500 000 000 to the faucet.
5	Where deploys go	A randomly selected active validator, redrawn at each start and after each resize; with G4, the whole list.
6	Timings	Tuned for a laptop (profile <code>laptop-v1</code> : casper loop 1.0 s at $N = 2$ rising to 3.0 s at $N = 10$; heartbeat on; maximum LFB age 15 s, or 30 s above $N = 4$); recalibrated by T1.
7	Foreign Origin	The node refuses it (N5).
8	Embers	An installation option, default on.

19.3 Resolved on 7 October 2026

Decisions 9–17 were opened by version 0.4 and implemented as proposed (see the implementation notes); they are kept here, numbered as before, as the record. Decision 11 now registers F1R3Ink too, its client having landed.

Open decision 9 (The F1R3Games port block). 40700–40709, beside `ign1t10n`’s other blocks and apart from a developer’s 8640, 5173 and 5174 (proposed); or the service’s own default, 8640, which would collide with a development portal.

Open decision 10 (Who is the Cooperative on the local shard). A key `ign1t10n` generates and holds (proposed): registration then needs nothing from the person, and the key governs nothing

outside this shard. Or the person's own key, imported, so that the same identity governs every shard they run.

Open decision 11 (Which games to register). Install all five environments; register only games whose web client is bundled, F1R3Pix and F1R3Beat (proposed), adding the others as clients land. Or register all five now, which lists three games the browser cannot play.

Open decision 12 (One origin per game). Per-game origins (proposed), as the portal's isolation model intends; or one shared games origin, which is simpler but lets games read one another's storage.

Open decision 13 (Opening the browser). At the end of first run, open the portal in the default browser as well as F1R3Gaze (proposed); or show only the "Open F1R3Games" button.

Open decision 14 (Faucet amount). 100 F1R3 per new portal key on the local shard (proposed), enough for allowances and F1R3Pix and F1R3Beat payments between players; or the service default of 1 F1R3.

Open decision 15 (The F1R3Beat breeder). Off by default, a switch in the panel (proposed), since a single person's local shard produces little engagement to select on; or on, daily.

Open decision 16 (Ship the CLI). Bundle `f1r3games` in `Helpers` with `ctl games env` printing how to point it at the local portal (proposed); it is how a developer scripts plays and runs `beat verify`.

Open decision 17 (F1R3Games by default). On by default in the first-run window, like `Embers` (proposed).

19.4 Open decisions

Opened by version 0.5. Each is implemented as proposed, as a setting or a constant, so a different answer changes a default, not the design.

Open decision 18 (The relay by default). On whenever F1R3Ink is installed (proposed, and implemented): a round that offers anonymous ink then works, and the relay's cost on a local shard is the faucet's. Or off, so that nothing on the Mac learns who inks whom until the person chooses it.

Open decision 19 (Where the relay runs). Inside the portal process, as `f1r3games-service` implements it (proposed, and implemented). Or a second `f1r3games-service` process serving only `/api/relay`, with its own port and the relay's keys alone, which isolates those keys from the portal but needs F10 and a manifest `relay` on another origin, and so CORS.

Open decision 20 (The relay's URL while it is off). F1R3Ink's manifest names the relay whether or not it runs (proposed, and implemented, because of Finding 23); off, the portal answers 404. With F9, the alternative is to register F1R3Ink without `relay` while it is off, so the shell says "no relay" at launch rather than at the first anonymous ink.

Open decision 21 (Upgrading a never-registered environment). Replace it without asking (proposed, and implemented), since no instance can exist; or treat it as any other upgrade and ask, which makes every version 0.4 install show an "F1R3Ink update" dialog that discards nothing.

Open decision 22 (Relay funding). 1 000 F1R3 at G3, topped up to 1 000 below 100 (proposed); or the faucet's remaining balance as the only limit. T3 measures `phlo` per batch to settle it.

Open decision 23 (Anonymous ink on a one-person shard). Keep F1R3Ink’s minimum of five players in a round for anonymous ink (proposed): a person alone on the local shard, with one browser, cannot use it, which is the point of the minimum. Or let a launch lower it on the local shard, for demonstrations; that is a launch option, not an installer setting.

A Configuration templates

A.1 Node configuration

conf/common.conf, bootstrap.conf, validator- $\{k\}$.conf and observer.conf are version 0.3’s, with the laptop-v1 timing values in place of docker/conf/default.conf’s and logging.sink = "stdout":

```
# conf/common.conf (excerpt): rendered by ignite10n; values for N validators
protocol-server { network-id = "{{NETWORK_ID}}", bind-address = "127.0.0.1", # N1
                  allow-private-addresses = true, no-upnp = true }
protocol-client.network-id = "{{NETWORK_ID}}"
peers-discovery { bind-address = "127.0.0.1", lookup-interval = 60 seconds }
api-server { host = "127.0.0.1", reject-foreign-origin = true } # N5
casper {
  shard-name = "root"
  fault-tolerance-threshold = 0.1
  casper-loop-interval = {{LOOP_MS}} ms # 1000 at N = 2 .. 3000 at N = 10
  heartbeat { enabled = true, max-lfb-age = {{LFB_AGE}} seconds }
  genesis-ceremony { required-signatures = {{max(N0-1,1)}},
                    approve-interval = 10 seconds, approve-duration = 1 minutes }
  genesis-block-data { bonds-file = "{{STATE}}/genesis/bonds.txt",
                      wallets-file = "{{STATE}}/genesis/wallets.txt",
                      number-of-active-validators = 10,
                      epoch-length = 10, quarantine-length = 10 }
}
logging { format = "json", sink = "stdout" }
```

A.2 games/f1r3games.toml

```
# Rendered by ignite10n at G4 and at every start (the validators are redrawn).
# games/f1r3games-jobs.toml is the same file without [relay].
listen = ["127.0.0.1:40700", "::1:40700"] # F4
public_host = "localhost:40700" # F4
shard_id = "root"
validator_url = "http://127.0.0.1:40420" # the first of validator_urls
validator_urls = ["http://127.0.0.1:40420", "http://127.0.0.1:40410"] # F7, shuffled
observer_url = "http://127.0.0.1:40453"
coop_address = "{{LOCAL_COOPERATIVE_ADDRESS}}"
env_version = 1 # from Resources/f1r3games/
  games.toml
phlo_price = 1
default_phlo_limit = 500000
max_phlo_limit = 50000000
env_phlo_limit = 50000000
deploy_ttl_ms = 600000
```

```

token_ttl_secs      = 3600
bootstrap_env       = false           # installation is a job (G4), not a side effect of
    serve
cors_origins        = []              # the shell is same-origin
portal_base_url     = "http://localhost:40700"
static_dir          = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/portal"
# keys come from the environment (F1); none is ever in this file

[faucet]
enabled = true
amount  = 100000000000               # 100 F1R3 (Decision 14)

[[origins]]                      # F2
id      = "f1r3pix"
listen  = ["127.0.0.1:40701", "::1:40701"]
public_host = "localhost:40701"
dir      = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/games"

[[origins]]
id      = "f1r3beat"
listen  = ["127.0.0.1:40702", "::1:40702"]
public_host = "localhost:40702"
dir      = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/games"

[[origins]]                      # v0.5
id      = "f1r3ink"
listen  = ["127.0.0.1:40703", "::1:40703"]
public_host = "localhost:40703"
dir      = "/Applications/ign1t10n.app/Contents/Resources/f1r3games/games"

[relay]                        # F8, v0.5: only in the portal's file, only when the
    relay runs
enabled      = true
base_url     = "http://localhost:40700/api/relay"
window_blocks = 3
per_hour     = 30

```

B The manifest

Schema 4 adds `[options].games`, `[options].games_open_at_first_run` and the `[games]` tables to schema 3; an absent `[games]` reads as F1R3Games never installed, so a schema-3 manifest upgrades in place. Version 0.5 adds three keys to `[games]` without changing the schema number: `relay` (absent reads as true), `relay_address` and `relay_named` (absent reads as not named, so a version 0.4 manifest names the relay at its first start).

```

# ~/Library/Application Support/F1R3FLY/ign1t10n/shard.toml (excerpt)
schema = 4

[options]
embers = true
gaze_integration = true
deploy_target = "random"

```

```

timing = "laptop-v1"
games = true
games_open_at_first_run = true

[games]
revision = "d2381c6"
portal_port = 40700
validator_slot = 2
service_address = "1111...Xs"
coop_address = "1111...Cp"
breeder_address = "1111...Br"
env_uri = "rho:id:...g7"
env_version = 1
faucet_f1r3 = 100
breeder = false
relay = true
    true)
relay_address = "1111...Rl"
relay_named = 2
    for

[[games.game]]
id = "f1r3pix"
port = 40701
env_uri = "rho:id:i77c...gtgn"
env_version = 2
client = true
registered = true
manifest_sha256 = "5e0c...91"

[[games.game]]
id = "f1r3beat"
port = 40702
env_uri = "rho:id:...q4"
env_version = 1
client = true
registered = true
manifest_sha256 = "a3b1...0d"

[[games.game]]
id = "f1r3ink"
port = 40703
env_uri = "rho:id:...k2"
env_version = 2
    never registered)
client = true
registered = true
manifest_sha256 = "77d2...4e"

[games.stages]
    is relay_named
secrets = true
ports = true
funded = true

```

the F1R3Games revision last installed

drawn at each start

v0.5: run F1R3Ink's relay (absent reads as

F1R3Ink's environment version it was named

1 under v0.4, replaced without consent (

G1..G6 and G8, cleared from G3 at reset; G7

```
portal_env = true
game_envs = true
registered = true
opened = true

[games.job]                                # present only while a job runs
kind = "register"
game = "fir3beat"
started = 2026-10-14T09:20:03Z
deploy = "3045...aa"
```